Chapter 6

Anatomy of Flowering Plants

Structural similarities and variations are easy to see in the external morphology of larger living organisms. Study the internal structure and one finds several similarities and differences there too.

What anatomy is

- the study of the internal structure of plants
- plants have cells as the basic unit, cells are organised into tissues, and tissues in turn into organs
- different organs in a plant differ in their internal structure, and within angiosperms the monocots and dicots are anatomically different as well
- internal structures also show adaptations to diverse environments

Tissues vary by where they sit

- tissues have been considered by the types of cells present; they also vary by their location in the plant body
- structure and function are both dependent on that location
- on the basis of structure and location there are three tissue systems

The three tissue systems

- the epidermal tissue system
- the ground, or fundamental, tissue system
- the vascular, or conducting, tissue system

In this chapter

6.2 The epidermal tissue system

6.3 The ground tissue system

6.4 The vascular tissue system

6.5.1 Dicotyledonous root

6.6.1 Dicotyledonous stem

6.7.1 Dorsiventral leaf

Guard cells, bean-shaped, enclosing the stomatal pore

Guard cells, dumb-bell shaped, as in grasses

Subsidiary cells, epidermal cells specialised around the guard cells

Chloroplasts within the guard cells

Together, pore and cells make the stomatal apparatus

Figure 6.A Diagrammatic representation: (a) stomata with bean-shaped guard cells (b) stomata with dumb-bell shaped guard cell

6.2 The epidermal tissue system

The epidermis

- it forms the outer-most covering of the whole plant body, and comprises epidermal cells, stomata and the epidermal appendages, the trichomes and hairs
- the outermost layer of the primary plant body, made of elongated, compactly arranged cells forming a continuous layer, and usually single-layered
- epidermal cells are parenchymatous, with a small amount of cytoplasm lining the cell wall and a large vacuole
- the outside is often covered by a waxy thick cuticle which prevents the loss of water, though cuticle is absent in roots

Stomata

- structures present in the epidermis of leaves, regulating transpiration and gaseous exchange
- each stoma is composed of two bean-shaped guard cells enclosing the stomatal pore, and in grasses the guard cells are dumb-bell shaped
- the outer walls of the guard cells, away from the pore, are thin, while the inner walls towards the pore are highly thickened
- the guard cells possess chloroplasts and regulate the opening and closing of the stomata

Subsidiary cells and hairs

- a few epidermal cells near the guard cells may become specialised in shape and size, and are known as subsidiary cells
- the stomatal aperture, guard cells and surrounding subsidiary cells together are the stomatal apparatus
- root hairs are unicellular elongations of the epidermal cells that help absorb water and minerals from the soil
- on the stem the epidermal hairs are trichomes, usually multicellular, branched or unbranched, soft or stiff, sometimes secretory, and they help prevent water loss by transpiration

6.3 The ground tissue system

Everything but epidermis and vascular bundles

- it consists of the simple tissues parenchyma, collenchyma and sclerenchyma
- parenchymatous cells are usually present in cortex, pericycle, pith and medullary rays, in the primary stems and roots
- in leaves the ground tissue is thin-walled chloroplast containing cells, and is called mesophyll

Radial, xylem and phloem on alternate radii, as in roots

Conjoint closed, xylem and phloem on the same radius, no cambium

Conjoint open, the same but with cambium between them

Figure 6.B Various types of vascular bundles : (a) radial (b) conjoint closed (c) conjoint open

6.4 The vascular tissue system

Open and closed bundles

- the vascular system consists of the complex tissues phloem and xylem, which together constitute vascular bundles
- in dicotyledonous stems cambium is present between phloem and xylem, so these bundles can form secondary xylem and phloem and are called open
- in monocotyledons no cambium is present, so no secondary tissues form and the bundles are referred to as closed

Radial and conjoint

- when xylem and phloem within a bundle are arranged alternately along different radii the arrangement is radial, as in roots
- in the conjoint type xylem and phloem lie together along the same radius, common in stems and leaves
- conjoint bundles usually have the phloem located only on the outer side of the xylem

6.5.1 The dicotyledonous root

From the outside in

- for a better understanding of the tissue organisation of roots, stems and leaves it is convenient to study transverse sections of the mature zones of these organs
- the outermost layer is the epiblema, many of whose cells protrude as unicellular root hairs
- the cortex consists of several layers of thin-walled parenchyma cells with intercellular spaces

Epidermis with root hairs on the outside

Cortex, then endodermis with its casparian strips

Pericycle inside the endodermis

Protoxylem and metaxylem, with phloem between the arms

Pith at the centre, small in dicot and large in monocot

Figure 6.C T.S. : (a) Dicot root (Primary) (b) Monocot root

Endodermis and pericycle

- the innermost layer of the cortex is the endodermis, a single layer of barrel-shaped cells without intercellular spaces
- the tangential and radial walls of the endodermal cells carry a deposition of the water-impermeable waxy material suberin, in the form of casparian strips
- next to the endodermis lie a few layers of thick-walled parenchymatous cells, the pericycle, where lateral roots and the vascular cambium of secondary growth are initiated

The stele

- the pith is small or inconspicuous
- the parenchymatous cells lying between xylem and phloem are called conjuctive tissue
- there are usually two to four xylem and phloem patches, and later a cambium ring develops between them
- all tissues on the inner side of the endodermis, that is pericycle, vascular bundles and pith, constitute the stele

6.5.2 The monocotyledonous root

Similar, with four differences

- the anatomy is similar to the dicot root in many respects, having epidermis, cortex, endodermis, pericycle, vascular bundles and pith
- where the dicot root has fewer xylem bundles, the monocot root usually has more than six, that is polyarch
- the pith is large and well developed
- monocotyledonous roots do not undergo any secondary growth

Dicot, vascular bundles in a ring, conjoint and open

Monocot, bundles scattered through the ground tissue, conjoint and closed

Figure 6.D T.S. of stem : (a) Dicot (b) Monocot

6.6.1 The dicotyledonous stem

Epidermis and cortex

- the epidermis is the outermost protective layer of the stem, covered with a thin layer of cuticle, and it may bear trichomes and a few stomata
- the cells arranged in multiple layers between epidermis and pericycle constitute the cortex, which consists of three sub-zones

The three sub-zones of the cortex

- the outer hypodermis, a few layers of collenchymatous cells just below the epidermis, giving mechanical strength to the young stem
- cortical layers below the hypodermis, of rounded thin walled parenchymatous cells with conspicuous intercellular spaces
- the innermost layer, the endodermis, whose cells are rich in starch grains, so that the layer is also referred to as the starch sheath

Dicot, epidermis with cuticle, mesophyll of palisade and spongy parenchyma

Monocot, stomata on both surfaces and mesophyll not differentiated

Vascular bundles in the veins and midrib, within bundle sheath cells

Figure 6.E T.S. of leaf : (a) Dicot (b) Monocot

Pericycle, rays and pith

- the pericycle lies inside the endodermis and above the phloem, as semi-lunar patches of sclerenchyma
- between the vascular bundles are a few layers of radially placed parenchymatous cells, the medullary rays
- a large number of vascular bundles are arranged in a ring, and that ring arrangement is a characteristic of the dicot stem; each bundle is conjoint, open, and with endarch protoxylem
- a large number of rounded parenchymatous cells with large intercellular spaces occupy the central portion and constitute the pith

6.6.2 The monocotyledonous stem

Scattered bundles

- a sclerenchymatous hypodermis, and a large number of scattered vascular bundles each surrounded by a sclerenchymatous bundle sheath
- a large, conspicuous parenchymatous ground tissue
- vascular bundles are conjoint and closed, and the peripheral ones are generally smaller than the centrally located ones
- the phloem parenchyma is absent, and water-containing cavities are present within the bundles

6.7.1 The dorsiventral leaf

Epidermis and mesophyll

- the vertical section through the lamina shows three main parts: epidermis, mesophyll and vascular system
- the epidermis covers both the upper adaxial and lower abaxial surfaces and has a conspicuous cuticle; the abaxial epidermis generally bears more stomata than the adaxial, which may even lack them
- the tissue between the two epidermes is the mesophyll, which possesses chloroplasts, carries out photosynthesis and is made of parenchyma
- it has two types of cells, the palisade parenchyma and the spongy parenchyma, the adaxially placed palisade being elongated cells arranged vertically and parallel to each other

Spongy parenchyma and the veins

- the oval or round and loosely arranged spongy parenchyma sits below the palisade cells and extends to the lower epidermis, with numerous large spaces and air cavities between the cells
- the vascular system includes the vascular bundles seen in the veins and the midrib, their size depending on the size of the veins
- the veins vary in thickness in the reticulate venation of dicot leaves
- the vascular bundles are surrounded by a layer of thick walled bundle sheath cells

6.7.2 The isobilateral leaf

Two differences from the dorsiventral leaf

- the stomata are present on both surfaces of the epidermis
- the mesophyll is not differentiated into palisade and spongy parenchyma
- the parallel venation of monocot leaves shows as near similar sizes of vascular bundles, except in the main veins, in vertical sections

Bulliform cells

- in grasses certain adaxial epidermal cells along the veins modify into large, empty, colourless cells
- when they have absorbed water and are turgid, the leaf surface is exposed
- when they are flaccid from water stress they make the leaves curl inwards, to minimise water loss

6.8 Summary

Tissues and tissue systems

- plant tissues are broadly classified into meristematic, that is apical, lateral and intercalary, and permanent, that is simple and complex
- assimilation of food and its storage, transport of water, minerals and photosynthates, and mechanical support are the main functions of tissues
- the epidermal system is made of epidermal cells, stomata and the epidermal appendages; the ground system forms the main bulk of the plant and divides into cortex, pericycle and pith
- the vascular system is formed by xylem and phloem, the bundles differing by presence of cambium and location of xylem and phloem, and they translocate water, minerals and food

Monocot against dicot

- monocotyledonous and dicotyledonous plants show marked variation in their internal structures
- they differ in type, number and location of vascular bundles
- secondary growth occurs in most dicotyledonous roots and stems

From this chapter

1. Draw illustrations to bring out the anatomical difference between (a) monocot root and dicot root (b) monocot stem and dicot stem
2. Cut a transverse section of the young stem of a plant from your school garden and observe it under the microscope. How would you ascertain whether it is a monocot stem or a dicot stem? Give reasons.
3. The transverse section of a plant material shows these anatomical features: (a) the vascular bundles are conjoint, scattered and surrounded by sclerenchymatous bundle sheaths (b) phloem parenchyma is absent. What will you identify it as?
4. What is a stomatal apparatus? Explain the structure of stomata with a labelled diagram.
5. Name the three basic tissue systems in flowering plants. Give the tissue names under each system.
6. How is the study of plant anatomy useful to us?
7. Describe the internal structure of a dorsiventral leaf with the help of labelled diagrams.